Book 1 — Lecture 2 — Exercise 1

Derivatives of Four Introductory Functions

Prompt. Calculate the derivatives[1] of each of these functions:

$$\begin{aligned} f(t) &= t^4 + 3t^3 - 12t^2 + t - 6, \\ g(x) &= \sin(x) - \cos(x), \\ \theta(\alpha) &= e^\alpha + \alpha \ln(\alpha), \\ x(t) &= \sin^2(t) - \cos(t). \end{aligned}$$

$f(t)$ (polynomial)

Differentiate term by term using $\dfrac{d}{dt} t^n = nt^{n-1}$:[2]

$$f'(t) = 4t^3 + 9t^2 - 24t + 1.$$

$g(x)$ (sine and cosine)

Using $(\sin x)' = \cos x$ and $(\cos x)' = -\sin x$,[3]

$$g'(x) = \cos(x) + \sin(x).$$

$\theta(\alpha)$ (exponential plus $\alpha \ln \alpha$)

Here $\alpha > 0$ so that $\ln(\alpha)$ is defined.[4] We have $(e^\alpha)' = e^\alpha$,[5] and by the product rule,[6]

$$\frac{d}{d\alpha}(\alpha \ln(\alpha)) = 1 \cdot \ln(\alpha) + \alpha \cdot \frac{1}{\alpha} = \ln(\alpha) + 1.$$

Therefore

$$\theta'(\alpha) = e^\alpha + \ln(\alpha) + 1.$$

[1] A *derivative* measures how sensitive a function's output is to a small change in its input; on a graph, it is the slope of the tangent line at a point. The prime notation f' means "the derivative of f with respect to the variable named in context."

[2] This is the *power rule*. Sums and constant multiples of terms differentiate term-by-term (*linearity* of the derivative operator).

[3] These are the basic trigonometric derivatives with respect to the angle x (measured in radians, as is standard in calculus and physics).

[4] ln is the natural logarithm (base e). It requires a positive argument $\alpha > 0$.

[5] The exponential e^x equals its own derivative: $\frac{d}{dx} e^x = e^x$.

[6] The *product rule*: if u and v are differentiable functions of the same variable, then $(uv)' = u'v + uv'$.

$x(t)$ (chain rule on $\sin^2 t$)

Write $\sin^2(t) = (\sin t)^2$. By the chain rule,[7]

$$\frac{d}{dt}\sin^2(t) = 2\sin(t)\cos(t) = \sin(2t).$$

[8] Also $\dfrac{d}{dt}(-\cos t) = \sin t$.[9] Hence

$$x'(t) = 2\sin(t)\cos(t) + \sin(t) = \sin(t)\big(2\cos(t) + 1\big).$$

Result. Summarizing,

$$\begin{aligned} f'(t) &= 4t^3 + 9t^2 - 24t + 1, \\ g'(x) &= \cos(x) + \sin(x), \\ \theta'(\alpha) &= e^{\alpha} + \ln(\alpha) + 1 \quad (\alpha > 0), \\ x'(t) &= \sin(t)\big(2\cos(t) + 1\big). \end{aligned}$$

Example. At $\alpha = 1$: $\theta'(1) = e^1 + \ln(1) + 1 = e + 1$, since $\ln(1) = 0$.

[7] The *chain rule* composes rates of change: if $y = h(u)$ and $u = k(t)$, then $\frac{dy}{dt} = h'(u)\,k'(t)$. Here $h(u) = u^2$ and $u = \sin t$.

[8] The identity $2\sin t\cos t = \sin(2t)$ is the sine double-angle formula.

[9] Constants factor out of derivatives, and $(\cos t)' = -\sin t$, so $\frac{d}{dt}(-\cos t) = -(-\sin t) = \sin t$.